AFRILANG Stdlib

std/c/mlkmeans

Bibliothèque AFRILANG `std/c/mlkmeans` (niveau complex, catégorie complex). Elle expose 6 fonction(s) : kmeans, assignCl

`import "std/c/mlkmeans"` · `use mlkmeans`

- `kmeans(points list number, k number, dims number) ? list number`
- `assignClusters(points list number, centroids list number, dims number) ? list number`
- `inertia(points list number, centroids list number, dims number) ? number`
- `clusterSizes(labels list number, k number) ? list number`
- `silhouetteSample(points list number, labels list number, idx number, dims number) ? number`
- `elbowK(points list number, maxK number, dims number) ? number`

Category: complex | Tier: complex | Functions: 6

FRANCAIS

1. Vue d'ensemble

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- `silhouetteSample(points list number, labels list number, idx number, dims number) ? number`
- `elbowK(points list number, maxK number, dims number) ? number`

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/mlkmeans"
use mlkmeans
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés ? graphes, NLP, réseaux complexes.

3. Référence API

? kmeans(points list number, k number, dims number) ? list
? assignClusters(points list number, centroids list number, dims number) ? list
? inertia(points list number, centroids list number, dims number) ? number
? clusterSizes(labels list number, k number) ? list
? silhouetteSample(points list number, labels list number, idx number, dims number) ? number
? elbowK(points list number, maxK number, dims number) ? number

4. Exemples d'utilisation

```
import "std/c/mlkmeans"
use mlkmeans

create result = kmeans(/* args */)
say result

create result = assignClusters(/* args */)
say result

create result = inertia(/* args */)
say result

create result = clusterSizes(/* args */)
say result

create result = silhouetteSample(/* args */)
say result

create result = elbowK(/* args */)
say result
```

5. Bonnes pratiques

? Préférez `import "std/c/mlkmeans"` en tête de fichier.
? Lancez `afrilang check` avant de livrer.
? Combinez avec les modules stdlib de la même catégorie.
? Voir /stdlib/ pour le source live et les démos playground.
? Signalez les problèmes sur GitHub si une export manque.

6. Annexe ? source

```
module mlkmeans

function kmeans(points list number, k number, dims number) returns list number
end
```

```
function assignClusters(points list number, centroids list number, dims number) returns list number
end

function inertia(points list number, centroids list number, dims number) returns number
end

function clusterSizes(labels list number, k number) returns list number
end

function silhouetteSample(points list number, labels list number, idx number, dims number) returns number
end

function elbowK(points list number, maxK number, dims number) returns number
end
end
```

ENGLISH

1. Overview

AFRILANG library `std/c/mlkmeans` (tier complex, category complex). Exports 6 function(s): kmeans, assignClusters, inert

`import "std/c/mlkmeans"` · `use mlkmeans`

- `kmeans(points list number, k number, dims number) ? list number`
- `assignClusters(points list number, centroids list number, dims number) ? list number`
- `inertia(points list number, centroids list number, dims number) ? number`
- `clusterSizes(labels list number, k number) ? list number`
- `silhouetteSample(points list number, labels list number, idx number, dims number) ? number`
- `elbowK(points list number, maxK number, dims number) ? number`

2. Installation & import

Add the module to your program:

```
import "std/c/mlkmeans"
use mlkmeans
```

Tier: complex · Category: Complex modules (c/)
Advanced domains ? graphs, NLP, complex networks.

3. API reference

? kmeans(points list number, k number, dims number) ? list
? assignClusters(points list number, centroids list number, dims number) ? list
? inertia(points list number, centroids list number, dims number) ? number
? clusterSizes(labels list number, k number) ? list
? silhouetteSample(points list number, labels list number, idx number, dims number) ? number
? elbowK(points list number, maxK number, dims number) ? number

4. Usage examples

```
import "std/c/mlkmeans"
use mlkmeans

create result = kmeans(/* args */)
say result

create result = assignClusters(/* args */)
say result

create result = inertia(/* args */)
say result

create result = clusterSizes(/* args */)
say result

create result = silhouetteSample(/* args */)
say result

create result = elbowK(/* args */)
say result
```

5. Best practices

? Prefer `import "std/c/mlkmeans"` at the top of your file.
? Run `afrilang check` before shipping.
? Combine with related stdlib modules from the same category.
? See /stdlib/ for the live source and playground demos.
? Report issues on GitHub if an export is missing or undocumented.

6. Appendix ? source

```
module mlkmeans

function kmeans(points list number, k number, dims number) returns list number
end
```

```
function assignClusters(points list number, centroids list number, dims number) returns list number
end
function inertia(points list number, centroids list number, dims number) returns number
end
function clusterSizes(labels list number, k number) returns list number
end
function silhouetteSample(points list number, labels list number, idx number, dims number) returns number
end
function elbowK(points list number, maxK number, dims number) returns number
end
end
```